



Institute of Infrastructure Technology Research and Management

End Semester
(Start time: 09.30 p.m. and End time: 12:30 p.m.)
B. Tech. (All Branches)
Semester - VI

Date: 24/04/2024

Course Name: Infrastructure Planning and Management

Course Code : HS 225002

Max. Marks: 60

Instructions:

1. Course instructor may penalize the student by deducting the marks, if answer is adjudged to be copied during assessment. This act shall be recorded as UFM (unfair means).
2. Assume suitable data wherever essential and mention it clearly.
3. Writing appropriate units, nomenclature, and drawing neat sketches/schematics wherever required is an integral part of the answer.

Section A

Attempt all the questions. Each question carries 10 marks. (10X 3 = 30 Marks)

- Q.1) Critically examine the primary issues in infrastructure development using the Dabhol power plant case as a reference.
- Q.2) Why is the government following a PPP model in spite of the large viability gap funding requirements? Explain your answer with the help of Vizhinjam International Seaport Ltd (VISL) case.
- Q.3) Discuss the various infrastructure risks and mitigation strategies associated with Aviation sector. Critically discuss the AirAsia India 2017 case.

Section B

Attempt all the questions. Each question carries 10 marks. (10 X 3 = 30 Marks)

- Q.1) L & T infrastructure is planning to undertake a bullet train project from Delhi to Ahmedabad. The cash flows for this project are provided below. Determine the Net Present Value (NPV) of the project and advise whether L&T Infrastructure should proceed with it or not. Assume a discount rate of 10%. The initial cash outlay is Rs. 1,00,000.

Year	Cash flow
2024	20000
2025	25000
2026	30000
2027	35000
2028	20000



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Q.2) Utilize the Net Present Value (NPV) method to compare and analyse which proposal is superior to accept. Assume a discount rate of 10%.

Project	Cash outflow	Cash Inflow		
	Year 0	Year 1	Year 2	Year 3
X	20000	30000	20000	10000
Y	20000	10000	20000	30000

Q.3) Adani infrastructure is planning to build a power plant which has the following projections for unit sales

Year	1	2	3	4	5
Unit Sales	50000	100000	100000	70000	50000
Price Per Unit	150	150	150	150	130

In this project following are the financials

- Cost of new plant, equipment and installation – 10 million
- Variable Cost – 80/Unit . Fixed Costs – 5,00,000 per year
- Working Capital – 100000 to get production started. For each year investment will be 10% of sales in the year
- Straight line depreciation, with no salvage value
- Tax Rate – 34%
- Cost of Capital = 15%-

Considering the provided information, compute the distinct cash flows and recommend whether this project is financially feasible or not based on NPV calculation.